

Product Requirements Document (PRD): POS & Replenishment Backlog

Project Name: Star Convenience Store (SCS) Digital Transformation

System Name: SCS Order Management System (OMS)

Document Owner: Ritvik Sakpal (Business Analyst)

Document Control:

Metadata Field	Details
Document ID	PRD-SCS-OMS-001
Current Status	Draft - Under Review
Version	0.4
Created Date	August 19, 2026
Last Updated	August 21, 2026
Related Documents	SCS Business Requirements Document (BRD) v4.0, SCS Use Case Specification

Revision History:

Version	Date	Author	Description of Changes
0.1	Aug 19, 2026	Ritvik Sakpal	Initial draft of Epic 1 (POS Checkout: Features 1-5)
0.2	Aug 20, 2026	Ritvik Sakpal	Added Epic 2 (Stock Alerts & Replenishment: Features 6-7)
0.3	Aug 20, 2026	Ritvik Sakpal	Drafted Features 8-10. Refined Checkout exception logic.
0.4	Aug 21, 2026	Ritvik Sakpal	Consolidated all 10 features, applied standard Gherkin syntax, and integrated partial delivery/damaged goods scenarios.

Feature 1: POS Barcode Scanning

As a Retail Associate,

I want to scan a product barcode at the checkout counter

So that the system instantly retrieves its name, pricing, and tax details from the database and adds it to the transaction, reducing manual writing errors and cutting customer checkout times

Scenario 1: Barcode is scannable

- **Given** the barcode scanner is connected and online
- **And** the product master table is up-to-date with "Lays Chips" (EAN: 89012345)
- **When** the associate scans the product barcode "89012345"
- **Then** the system should decode the barcode and query the Product Master
- **And** retrieve the matching item's details (Product ID/EAN, Product Name, Product Unit Price, Product Tax %, and Product Total Price)
- **And** display the itemized line on the POS checkout screen

Scenario 2: Barcode is not scannable

- **Given** the barcode scanner is connected and online
- **When** the associate attempts to scan a physical barcode that is dirty, torn, or unreadable
- **Then** the system should notify the associate of the scan failure
- **And** provide an alternative manual input field on the interface to type in the Product ID or Product Name
- **And** search the Product Master upon selection to fetch and display the product details (Product ID, Product Name, Product Unit Price, Product Tax %, and Product Total Price)

Feature 2: Variable-Weight Scale Support

As a Retail Associate,

I want to search for a loose product by its Name or ID and read its weight directly from the integrated digital scale,

So that the system automatically calculates the correct total price based on the unit-weight rate, preventing manual entry errors and speeding up checkout times

Scenario 1: Calculation for Loose/Variable-Weight Items

- **Given** the digital weight scale is connected and online
- **And** the Product Master is up-to-date with "Onions" configured at 50 INR/KG
- **When** the Retail Associate places onions on the physical scale
- **And** searches and selects "Onions" (via Name or Product ID/EAN) on the POS terminal
- **Then** the system should dynamically read the stable weight of 500 grams (0.5 kg) from the scale
- **And** calculate the correct line item price of 25 INR ($50 \text{ INR/KG} \times 0.5 \text{ kg}$)
- **And** add "Onions" to the active order list with the following itemized details displayed | Product ID | Product Name | Weight (kg) | Unit Rate (per kg) | Line Subtotal | Applied Tax (GST) | Total Line Price |

Scenario 2: Scale Disconnection / Communication Failure

- **Given** the Retail Associate has searched and selected a variable-weight item by Name or ID
- **When** the system attempts to poll the scale but the connection times out or fails
- **Then** the system should display a "Scale Unresponsive" warning prompt on the screen
- **And** provide an alternative manual input field for the Associate to type in the

verified weight to proceed with the transaction.

Feature 3: Manual Product Lookup

As a Retail Associate,

I want to manually search for a product by its Name or Product ID/EAN and add a custom quantity to the transaction,

So that I can seamlessly checkout non-barcoded countable items (such as eggs or lemons) and complete sales when physical barcodes are dirty, torn, or unreadable

Scenario 1: Manual Search

- **Given** the Retail Associate is on the active billing screen
- **When** the Associate manually types "Parle-G" into the product search bar
- **And** selects the correct "Parle-G" item from the matching search results
- **Then** the system should add the selected product to the active order list
- **And** automatically populate its Product ID, Unit Price, and a default quantity of 1.

Scenario 2: Adding Loose, Countable Non-Barcoded Items

- **Given** a customer brings 6 loose "Eggs" (which lack barcodes) to the checkout counter
- **And** "Eggs" are configured in the Product Master as a countable unit item at 6 INR per unit
- **When** the Associate searches and selects "Eggs" by Name or ID
- **And** inputs 6 as the quantity
- **Then** the system should calculate the line total as 36 INR
- **And** display "Eggs | Qty: 6 | Price: 36 INR" on the POS screen.

Scenario 3: Product Search Fails to Find Match

- **Given** the Associate is on the manual search screen
- **When** the Associate enters an item name or ID that does not exist in the Product Master
- **Then** the system should display a clear "No matching products found" alert
- **And** prompt the Associate to check the spelling or manually enter an alternative product

Feature 4: Multi-Payment Type Logging

As a Retail Associate,

I want to select and log the customer's chosen payment method (Cash, Digital(Card/UPI), or Credit) during the checkout process,

So that the system accurately records the transaction type, facilitates the correct payment flow, and immediately triggers real-time stock deductions regardless of how the sale was settled.

Scenario 1: Transaction settled via Cash

- **Given** the POS terminal is at the payment selection screen for an active transaction totalling 150 INR
- **When** the Retail Associate selects "Cash" as the payment method
- **And** inputs a cash received amount of 200 INR
- **Then** the system should calculate and display the change due as 50 INR (200 INR received - 150 INR total)
- **And** allow the Associate to mark the transaction as "Paid - Cash"
- **And** complete the transaction, prompting the system to deduct inventory in real-time and print the customer receipt.

Scenario 2: Transaction settled via Digital Gateway

- **Given** the POS terminal is at the payment selection screen for an active transaction totalling 300 INR
- **When** the Retail Associate selects "Digital [Card/UPI]" as the payment method
- **And** initiates the external payment gateway on the POS terminal
- **Then** the system should wait for a successful payment response from the gateway
- **And** upon receiving success confirmation, mark the transaction as "Paid - Digital"

[Card/UPI]"

- **And** complete the transaction, prompting the system to deduct inventory in real-time and print the customer receipt.

Scenario 3: Transaction settled via Store Credit / Udhaar

- **Given** the customer is pre-authorized by the Store Owner to use store credit
- **And** the POS terminal is at the payment selection screen for an active transaction totalling 120 INR
- **When** the Retail Associate selects "Credit" as the payment method
- **Then** the system should mark the order status as "Paid - Credit"
- **And** generate a transaction receipt containing a unique Order ID
- **And** display an on-screen prompt instructing the Associate: "Please record Order ID [ID] in the physical Customer Credit Book"
- **And** complete the transaction to successfully decrement the physical inventory levels in real-time.

Feature 5: Receipt Generation

As a Retail Associate,

I want the system to automatically generate and print an itemized receipt with a unique Order ID, timestamps, tax breakdowns, and final totals upon payment confirmation,

So that I can hand a formal proof of purchase to the customer and ensure the transaction is permanently logged in the database for financial tracking.

Scenario 1: Successful Receipt Generation for Cash/Digital Sales

- **Given** the POS terminal is connected to the database and the receipt printer is online
- **And** the customer has completed payment of 150 INR via "Digital [UPI]"
- **When** the transaction is marked as completed in the system
- **Then** the system must generate a unique, sequential Order ID (e.g., SCS-2026-0819-001)
- **And** compile a receipt containing the following details:
 - **Store Header:** Star Convenience Store (SCS)
 - **Metadata:** Unique Order ID and Timestamp (Date/Time)
 - **Itemized List:** Product Name, Quantity/Weight, Unit Rate, and Line Total
 - **Financial Summary:** Subtotal (pre-tax), Applied GST (Tax %), Total Tax Amount, and Net Bill Total
 - **Payment Details:** Payment Method used (e.g., "Paid - Digital [UPI]")
- **And** trigger the printer to print the receipt
- **And** permanently write the complete transaction record to the Orders database table.

Scenario 2: Receipt Generation for Store Credit / Udhaar Transactions

- **Given** the customer is authorized to pay via "Credit"
- **And** the transaction total is 120 INR
- **When** the Associate completes the transaction using the "Credit" payment type
- **Then** the system must generate a receipt with a unique Order ID
- **And** mark the payment method on the receipt clearly as "Paid - Credit (Udhaar)"
- **And** display an on-screen prompt on the POS terminal: "Please manually record Order ID [ID] in the physical Customer Credit Book"
- **And** complete the transaction, prompting the system to decrement the physical inventory levels in real-time.

Scenario 3: Receipt Printing Exception / Hardware Error

- **Given** a transaction has been successfully paid and committed to the database
- **When** the receipt printer is out of paper or disconnected during printing
- **Then** the system must display an on-screen warning alert: "Printer Error: Receipt printing failed (Paper Out / Offline)"
- **And** ensure the transaction is not aborted or rolled back in the database
- **And** provide a prominent "Reprint Receipt" button on the POS interface, allowing the Associate to trigger a reprint once the printer hardware issue is resolved.

Feature 6: Real-Time Stock Auto-Deductions

As a Store Owner,

I want the system to automatically decrement a product's stock levels (by unit count or weight) in the Product Master immediately upon successful checkout,

So that my inventory is kept 100% accurate, allowing me to track active stock in real-time and prevent lost sales from unexpected stock outs.

Scenario 1: Order Completes

- **Given** the POS terminal is in active billing mode
- **And** the Product Master shows an initial stock of 50 units for the selected item
- **When** a retail associate scans and processes a checkout of 2 units of the item
- **And** settles the payment using the <Payment Type>
- **Then** the system should mark the order status as <Order Status>
- **And** immediately update the Product Master database to show the remaining stock as 48 units
- Examples:

Payment Type	Order Status
Cash	Paid - Cash
Digital (UPI)	Paid - Digital/UPI
Digital (Card)	Paid - Digital/Card
Credit	Paid - Credit

Scenario 2: Order fails

- **Given** the POS terminal is in active billing mode
- **And** the Product Master shows an initial stock of 50 units for the selected item
- **And** the retail associate has initiated the checkout of 2 units of the item
- **When** the transaction fails to complete due to <Failure Reason>
- **Then** the system should update the order status to <Order Status>
- **And** display a clear on-screen error message with buttons prompting the associate to either retry payment or cancel the order
- **And** skip the stock deduction step, keeping the Product Master count at 50 units
- Examples:

Failure Reason	Order Status
Internet Connection issue at the store	Fail - Internal Connection
Payment gateway response timeout	Fail - Payment Gateway Error
OMS database connectivity error	Fail - Software Error
Unspecified hardware or system failure	Fail - Other

Feature 7: Automated Low-Stock Notifications

As a Store Owner,

I want the system to send me instant low-stock alerts via Whatsapp, Email, SMS, and my store dashboard the moment a product's inventory falls to or below its minimum threshold.

So that I have immediate visibility into what needs to be reordered and can quickly review auto-generated draft POs before we hit a stock out and lose sales.

Scenario 1: Successful Notification

- **Given** the Product Master has an active stock level of 26 units for "Fresh Bread"
- **And** the product's minimum reorder threshold is configured at 25 units
- **When** a customer successfully purchases 1 unit of "Fresh Bread" at the register
- **Then** the system should decrement the database stock to 25 units
- **And** identify that the inventory level has fallen to or below the minimum threshold
- **And** automatically dispatch an instant notification (via Email and SMS) containing the following details:

Field	Value
Product ID/EAN	[Product ID/EAN]
Product Name	"Fresh Bread"
Minimum Threshold	25 units
Current Stock	25 units
Timestamp	[Current Date/Time of Sale]

- And immediately queue the Draft PO generation workflow for this product's assigned distributor.

Scenario 2: Notification Dispatch Fails

- **Given** the Product Master has an active stock level of 26 units for "Fresh Bread"
- **And** the product's minimum reorder threshold is configured at 25 units
- **When** a customer successfully purchases 1 unit of "Fresh Bread", decrementing the stock to 25 units
- **And** the system fails to send the alert via <Notification Channel> due to an external service disruption
- **Then** the system should log the alert status as "Pending Retry" in the notification queue
- **And** display a fallback warning on the Store Owner's POS dashboard: "Low-Stock Alert Delivery Failed via <Notification Channel>"
- **And** proceed to successfully save the Draft PO in the database, ensuring the replenishment process is not blocked.
- **Examples:**

Notification Channel
Store Official WhatsApp (API Gateway Timeout)
SMS Gateway (Carrier Network Offline)
Email Server (SMTP Authentication Refused)

Feature 8: Automated Draft PO Generation

As a Store Owner,

I want the system to automatically compile all low-stock products into draft Purchase Orders, grouped and consolidated by their assigned distributors,

So that I don't have to spend hours manually drafting vendor orders, allowing me to easily review and refine pre-filled replenishment lists on my dashboard.

Scenario 1: Auto-Generation of Draft PO

- **Given** the POS terminal is in active billing mode
- **And** the Product Master shows an initial stock of 26 units for "Kurkure"
- **And** "Kurkure" has a minimum threshold of 25 units and a target max stock of 60 units
- **When** a checkout transaction decrements the stock of "Kurkure" to 25 units
- **Then** the system should automatically generate a Draft PO for "Kurkure's" assigned distributor with status "Pending Approval"
- **And** calculate the reorder quantity as 35 units (Target Max 60 - Current Stock 25)
- And compile the Draft PO record to show the following details:

Field	Description
Product ID/EAN	[Product ID/EAN]
Product Name	"Kurkure"
Unit Price	[Product Unit Price]
Line Total	[Reorder Qty x Unit Price]

- **And** display a low-stock alert on the Store Owner's dashboard and dispatch a notification via the owner's configured alerts.

Scenario 2: Manually Modifying an Active Draft PO

- **Given** the Store Owner is reviewing an automatically generated Draft PO on the dashboard
- **When** the Store Owner selects the option to edit the Draft PO
- **Then** the system should allow the owner to select and remove specific products from the list
- **And** allow the owner to manually edit the target stock quantities, dynamically recalculating the total PO cost in real-time
- **And** allow the owner to manually add additional products to the PO, provided the added items match the current PO's assigned distributor.

Scenario 3: Cancelling a Draft PO

- **Given** the Store Owner is in the editing view of a Draft PO
- **When** the Store Owner selects the option to cancel the PO
- **Then** the system should update the PO status to "Cancelled"
- **And** remove the PO from the active Drafts view and archive it in the Cancelled folder
- **And** provide a "Restore PO" option on the archived record.

Scenario 4: Restoring a Cancelled PO

- **Given** the Store Owner is viewing a previously cancelled PO in the Cancelled folder
- **When** the Store Owner selects the "Restore PO" option
- **Then** the system should restore the PO record and return it to the active Drafts list
- **And** revert the PO status to "Pending Approval" so it can be edited or processed.

Feature 9: Digital PO Transmission

As a Store Owner,

I want the system to transmit finalized Purchase Orders directly to distributors via their preferred communication channel (WhatsApp Business API, Email, or SMS),

So that I can instantly place replenishment orders without manual typing or communication delays, accelerating stock deliveries and preventing out-of-stock situations.

Scenario 1: Successful PO Transmission

- **Given** a Draft PO with the status "Pending Approval" is open on the Store Owner's dashboard
- **When** the Store Owner reviews the order details and clicks "Approve & Send"
- **Then** the system should update the PO status to "Sent"
- **And** instantly dispatch a structured message payload via <Preferred Channel> to the distributor containing:

Field	Description
PO Order ID	[Unique PO ID]
Delivery Location	Star Convenience Store
Estimated Total Cost	[PO Total Cost]
Itemized List	[Product Names, EANs, and Approved Quantities]

- **And** log the transmission timestamp and selected channel in the database.

- Examples:

Preferred Channel
WhatsApp Business API
Email
SMS

Scenario 2: Transmission Gateway Fails

- **Given** a Draft PO with the status "Pending Approval" is open on the Store Owner's dashboard
- **When** the Store Owner clicks "Approve & Send" but the communication gateway times out or fails to respond
- **Then** the system should keep the PO status as "Pending Approval"
- **And** display a prominent on-screen warning: "Transmission Failed: [Gateway Offline] - Please try again or select an alternative channel"
- **And** allow the Store Owner to immediately retry the transmission or choose a different channel.
- Examples:

Channels
WhatsApp Business API
Email
SMS

Feature 10: Goods Receipt Note (GRN) Verification

As a Store Owner or Retail Associate,

I want to physically verify and scan incoming distributor shipments against the original open Purchase Order before submitting a Goods Receipt Note (GRN),

So that the Product Master database is updated with accurate stock counts and active low-stock alerts are automatically cleared.

Scenario 1: Full Delivery Received and Verified

- **Given** an open PO with status "Sent" exists in the system for "A1 Distributors" containing "Lays Chips" with an ordered quantity of 40 units
- **And** the distributor delivery arrives at the store with a paper challan
- **When** the Retail Associate scans the barcodes of the delivered items to verify and confirm receipt of all 40 units
- **And** the Store Owner locates the matching PO in the system and clicks "Submit GRN"
- **Then** the system should update the PO status to "Completed"
- **And** immediately increase the stock level of "Lays Chips" in the Product Master by 40 units
- **And** automatically clear the active low-stock dashboard alert and SMS/Email/WhatsApp notification queue for "Lays Chips".

Scenario 2: Partial Delivery Handling (Alternative Flow)

- **Given** an open PO with status "Sent" exists in the system for "A1 Distributors" containing "Fresh Bread" with an ordered quantity of 20 units
- **When** the distributor delivers only 12 units of "Fresh Bread" due to a supplier shortage
- **And** the Retail Associate scans the barcodes and verifies a partial receipt of only 12 units
- **And** the Store Owner clicks "Submit GRN" for the partial delivery
- **Then** the system should update the Product Master stock level of "Fresh Bread" by exactly 12 units
- **And** mark the original PO status as "Completed - Shortage"
- **And** automatically generate a new follow-up PO for the remaining 8 units of "Fresh Bread" with status "Pending Approval".

Scenario 3: Delivery Verification with Damaged and Rejected Goods

- **Given** an open PO with status "Sent" exists for "A1 Distributors" containing "Fresh Bread" with an ordered quantity of 20 units
- **When** the distributor delivers 20 units of "Fresh Bread"
- **And** the Retail Associate identifies that 4 units of "Fresh Bread" are crushed or stale and rejects them
- **And** the Retail Associate scans and logs 16 units as "Accepted" and 4 units as "Rejected - Damaged"
- **And** the Store Owner clicks "Submit GRN"
- **Then** the system should increase the Product Master stock level of "Fresh Bread" by exactly 16 units
- **And** update the PO status to "Completed - Damaged Items"
- **And** automatically log a pending credit/refund claim of 4 units of "Fresh Bread" in the distributor's payment dashboard queue for manual reconciliation